MoMa-Kitchen: A 100K+ Benchmark for Affordance-Grounded Last-Mile Navigation in Mobile Manipulation

Anonymous Author

Abstract

Mobile manipulation requires not only navigating to a target object but also positioning the robot base so that the arm can feasibly execute a manipulation task. Existing navigation benchmarks, however, reward proximity to a goal and neglect the last-mile *constraints imposed by arm reachability, base height, and environmental clutter. We introduce MoMa-Kitchen, a large-scale benchmark of 127,343 episodes across 569 procedurally generated kitchen scenes designed to bridge this gap. Each episode provides first-person RGB-D observations, global and floor point clouds, and a dense* floor affordance map *encoding the continuous probability of manipulation success as a function of robot base pose. The benchmark spans 137 asset types (65 rigid, 48 articulated, 24 obstacles) and six distinct robot arms with diverse reach and height profiles, enabling evaluation across robot morphologies. To assess the task, we propose NavAff, a lightweight baseline that fuses a global scene point cloud with a floor point cloud via cross-attention and conditions predictions on robot morphology parameters to output per-point affordance scores. Experiments show that NavAff outperforms adapted 3D detection baselines (PointNet++, VoteNet, H3DNet) across all metrics, achieving a Pearson Correlation Coefficient (PCC) of 0.680 and a Top-1 Manipulation Success Rate (MSR) of 0.72. Ablation studies confirm the importance of visual–geometric fusion and robot-aware conditioning. We further demonstrate sim-to-real transfer of the learned affordance predictor in real kitchen environments using a commodity RGB-D sensor and open-vocabulary segmentation.*

1. Introduction

A central aspiration of embodied AI is to deploy mobile manipulators—robots that both navigate and interact with their surroundings—in unstructured domestic settings [9, 20, 42]. Progress toward this goal requires agents to master two tightly coupled sub-problems: navigating to a target object and positioning the base precisely enough that the arm can reach and manipulate it. While the first sub-problem has received substantial attention through large-scale navigation benchmarks [2, 7, 13, 24], the second—which we call the *last-mile navigation problem*—is largely neglected.

Figure 1. Conventional navigation methods typically prioritize reaching a target location but do not account for constraints affecting manipulation feasibility. MoMa-Kitchen provides dense floor affordance maps (bottom) tied to manipulation success, enabling robots to navigate to base poses that actually support grasping.

Classical navigation benchmarks measure success by whether an agent reaches a goal position within a fixed threshold [1]. This proximity-based criterion is ill-suited for manipulation: a base pose that is within threshold of the target may be occluded by furniture, kinematically infeasible for the mounted arm, or colliding with obstacles. As illustrated in Figure 1, even state-of-the-art navigation systems can guide a robot into a pose where no manipulation can succeed.

The core challenge is *affordance-grounded navigation*: predicting the set of floor positions from which a specific robot arm can feasibly grasp or interact with a target object, given scene geometry, robot morphology, and clutter. Solving this problem requires a benchmark that (i) provides dense ground-truth feasibility labels over the floor, (ii) accounts for diverse robot morphologies, and (iii) spans a large variety of scenes and target objects to enable generalizable learning.

We present MoMa-Kitchen, a benchmark specifically designed to address these needs. MoMa-Kitchen contains 127,343 episodes collected across 569 procedurally varied kitchen scenes simulated in BestMan [40], a PyBullet-based mobile manipulation platform. Each episode is anchored to a first-person RGB-D frame captured from an arm-mounted camera and is paired with a dense *floor affordance map*—a per-point continuous score over the floor point cloud in-

AI-Generated

dicating the probability that placing the robot base at that location results in a successful manipulation attempt (grasp or suction). Labels are generated by exhaustive collision-aware manipulation trials at sampled base poses, followed by nearest-neighbor Gaussian densification to form smooth, continuous maps.

The benchmark includes 137 asset categories sourced from PartNet-Mobility [21] and spans six robot arms (Panda, Flexiv, UR5e, xArm6, Realman, Elephant) with varying reach and base heights, inducing diverse kinematic constraints. To evaluate on this benchmark, we design NavAff, a multi-modal neural network that encodes a global point cloud annotated with target and obstacle indicators, a floor point cloud, and a two-dimensional robot descriptor (base height and arm reach) into dense affordance predictions. NavAff employs a cross-attention Visual Affordance Module (VAM) to fuse scene-level context into floor-level queries, enabling it to incorporate information about remote obstacles and kinematic limits.

Our main contributions are:

- **MoMa-Kitchen**: A 127K+ episode benchmark for affordance-grounded last-mile navigation, with dense floor affordance maps, multi-robot support, and fully automated, reproducible data collection.
- **Automated pipeline**: A scene generation and labeling pipeline combining procedural layout randomization, collision-aware manipulation trials, and Gaussian densification for continuous affordance supervision.
- **NavAff**: A lightweight baseline model that fuses 2D–3D cues and robot morphology via cross-attention, outperforming adapted 3D detection methods across RMSE, logMSE, PCC, cosine similarity, and MSR.
- **Sim-to-real transfer**: Demonstration that a simulator-trained NavAff produces plausible floor affordances in real kitchen environments using an Intel RealSense D435i and Grounded SAM [27].

2. Related Work

Embodied Navigation Benchmarks. A large body of work has developed benchmarks for navigation in simulated photo-realistic environments [5, 11, 26, 37]. Vision-and-Language Navigation (VLN) [2, 3, 14, 15, 24] focuses on following natural-language instructions to reach a goal. Object-goal navigation [4, 19, 33] requires finding and reaching specific object instances. Embodied question answering [7, 34] combines navigation and perception. None of these formulations explicitly optimize for manipulation feasibility at the destination pose; MoMa-Kitchen fills this gap by providing dense manipulation-feasibility labels over the floor.

Mobile Manipulation. Mobile manipulation has been studied under several frameworks: hierarchical reinforcement learning [16], whole-body control [8, 20], teleoperation with imitation learning [9, 35], and modular skill coordination [36, 43]. Open-vocabulary mobile manipulation [17, 18, 25, 42, 45] extends these to arbitrary object categories. A key limitation shared by most of these works is that base pose selection at the last mile is either hand-crafted or ignored, motivating our benchmark. Related work MoMa-Pos [28] also optimizes base placement but uses inverse reachability maps computed at inference, without large-scale affordance supervision. Navi2Gaze [46] uses VLMs to select poses but does not provide a training dataset for affordance prediction.

Affordance Prediction. Affordance prediction [31] asks what actions an agent can perform at or with an object. In grasping, affordances are typically predicted at the object surface [30, 32, 39, 41]. VoxPoser [10] constructs 3D value maps for manipulation using language models. In contrast, MoMa-Kitchen defines affordances over the *floor*, capturing robot base positions suitable for manipulation—a novel and practically important variant of the problem.

Task-Constrained Base Placement. Determining feasible base configurations for manipulation is related to inverse reachability analysis, which precomputes valid base regions by inverting end-effector reachability maps. Classical methods require known target pose and free-space assumptions that break down under kitchen clutter. Sampling-based motion planners handle obstacles but operate at inference time, introducing latency incompatible with real-time navigation. Object placement prediction [20, 42] focuses on surface placement for objects rather than robot base positioning, and success is measured by post-hoc task execution rather than a dense label map. MoMa-Kitchen is the first benchmark, to our knowledge, to cast floor base pose selection as a supervised regression problem with continuous affordance maps, providing large-scale training data for feedforward base pose prediction directly from RGB-D observations without test-time planning.

Sim-to-Real Transfer in Robot Learning. Transferring robot learning models from simulation to the real world is a longstanding challenge. Domain randomization of visual textures, lighting, and physics parameters has been widely used to narrow this gap for grasping and navigation. Geometric representations such as point clouds tend to exhibit a smaller visual domain gap than raw RGB images, making them natural candidates for zero-shot sim-to-real transfer. Several works have shown that grasp quality networks trained on synthetic depth data transfer to real sensors with minimal performance degradation. Our experiments are consistent with this finding: by grounding predictions in geometric point cloud inputs rather than appearance features, a NavAff

AI-Generated

model trained entirely in MoMa-Kitchen produces plausible affordance maps in real kitchens without any fine-tuning, using Grounded SAM [27] for open-vocabulary segmentation.

3D Point Cloud Methods. We build on the PointNet++ hierarchy [22] for hierarchical point cloud encoding and use multi-scale grouping for set abstraction. We compare against VoteNet [23] and H3DNet [44], which are 3D object detection networks adapted to produce per-floor-point affordance scores. The cross-attention module in NavAff follows the general transformer paradigm [29].

Foundation Models for Manipulation Planning. Recent works have explored using large language models (LLMs) and vision-language models (VLMs) as high-level planners for robot manipulation [10]. While these approaches excel at task decomposition and semantic grounding, they typically rely on downstream motion planners or hand-crafted heuristics for base pose selection. MoMa-Kitchen provides the large-scale supervised training data needed to replace such heuristics with a learned, feedforward affordance predictor, and NavAff is directly composable with LLM-based task planners that can invoke it as a low-level policy module for the last-mile positioning step.

Simulation Platforms and Assets. MoMa-Kitchen is built on BestMan [40], a modular mobile manipulation simulator with unified hardware APIs. We use PartNet-Mobility assets [21] for articulated objects in the kitchen environment, following the SAPIEN [38] ecosystem.

3. The MoMa-Kitchen Benchmark

3.1. Problem Formulation

We formulate last-mile navigation for mobile manipulation as *floor affordance prediction*. Given a first-person RGB-D observation $\mathcal{I}$, a global scene point cloud $\mathcal{P}_g$, a floor point cloud $\mathcal{P}_f$, and a robot descriptor $\mathbf{r}$ encoding base height and arm reach, the task is to predict a continuous affordance map $\mathcal{A} : \mathcal{P}_f \rightarrow [0, 1]$ where $\mathcal{A}(p)$ is the probability that placing the robot base at floor point p enables successful manipulation of the target object.

3.2. Scene Construction

Simulator. All scenes are simulated using BestMan [40], a PyBullet-based platform that provides unified APIs for mobile manipulation. Scenes consist of rectangular kitchen layouts populated with procedurally arranged appliances (refrigerators, ovens, dishwashers), furniture (tables, counters, cabinets), and free-standing objects.

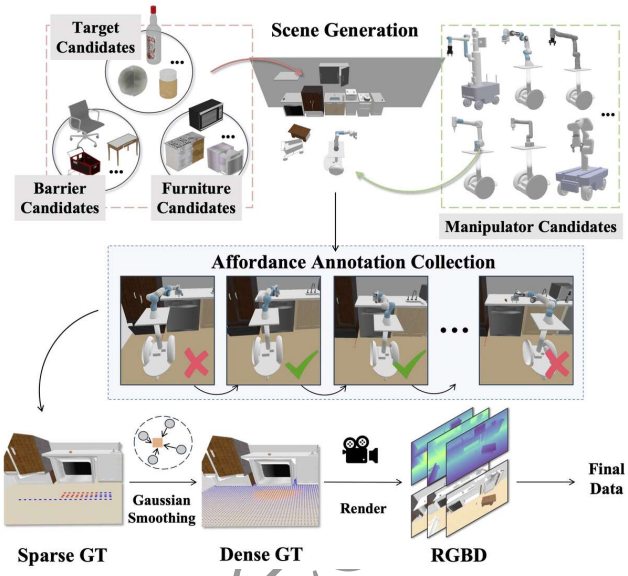

Figure 2. **Overview of scene setup and data generation pipeline.** Rectangular kitchen scenes are procedurally populated with appliances and furniture; targets and obstacles are placed randomly; the robot is equipped with an arm-mounted RGB-D camera. For each view, manipulation attempts are executed at sampled base poses and densified into continuous floor affordance maps.

Assets. Object assets are sourced from PartNet-Mobility [21] with additional rigid objects from ShapeNet [6]. The benchmark covers 137 asset types: 65 rigid objects (mugs, bottles, cans, tools, *etc.*), 48 articulated objects (drawers, doors, handles), and 24 obstacle categories (chairs, stools, carts, *etc.*). Asset instances are randomly selected and placed at diverse locations within each scene to maximize layout variety.

Robots. Six mobile manipulators are used: Franka Panda, Flexiv Rizon, Universal Robots UR5e, xArm6, Realman, and Elephant Robotics myCobot. As shown in Figure 3(a), these robots span a range of base heights (0.70–0.85 m) and maximum arm reaches (0.33–1.03 m), inducing distinct kinematic feasibility regions over the floor.

3.3. Data Collection Pipeline

View Sampling. For each target object and robot combination, 10 camera poses are selected per episode. Poses alternate left and right of the target, at increasing distances, until the target is visible in the RGB-D frame. This ensures diversity in viewing angle and distance while maintaining target visibility. Each pose provides an RGB image, a depth image, a global point cloud (scene), and a floor point cloud (height < 0.02 m).

AI-Generate

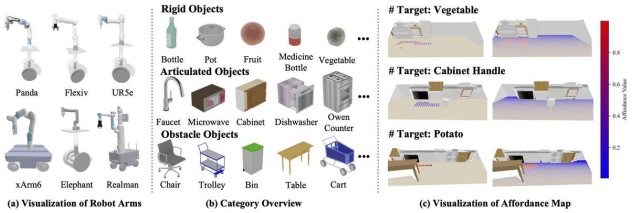

Figure 3. **(a) Robot arms used in MoMa-Kitchen.** Six manipulators with diverse base heights and arm reaches are employed. **(b)** Example floor affordance maps showing how feasible base regions (warm colors) vary with robot morphology and scene layout. **(c)** RGB-D views and associated point clouds demonstrate the first-person perspective used for prediction.

Affordance Sampling. For each episode, we define a semi-circular sampling region on the floor with radius equal to the arm's maximum reach, centered on the target. Base poses are sampled within this region at regular angular and radial intervals. At each sampled pose, the robot attempts manipulation (grasp or suction) of the target using collision-aware motion planning. The outcome (success/failure) constitutes the binary affordance label.

Frame Alignment and Densification. Binary success labels are transformed from world coordinates to the robot base frame and associated with the nearest neighbor in the floor point cloud. The resulting sparse binary labels are densified using k-nearest-neighbor Gaussian kernel interpolation:

$$\mathcal{A}(p) = \frac{\sum_{i \in \mathcal{N}_k(p)} \exp\left(-\frac{\|p-p_i\|^2}{2\sigma^2}\right) s_i}{\sum_{i \in \mathcal{N}_k(p)} \exp\left(-\frac{\|p-p_i\|^2}{2\sigma^2}\right)}, \quad (1)$$

where $s_i \in \{0, 1\}$ is the binary success label at neighbor p_i and σ is the Gaussian bandwidth. This produces a continuous, smooth affordance map over the entire floor point cloud, suitable for regression supervision.

Quality Control. Episodes are rejected if the target is not in line-of-sight from the camera, ensuring that the observation contains sufficient information about the target. Obstacle placement is validated to produce at least partial occlusion of the optimal approach region, ensuring non-trivial affordance maps. Camera intrinsics and all coordinate transforms are kept consistent across episodes.

3.4. Dataset Statistics

Table 1 summarizes the dataset split. MoMa-Kitchen contains a total of 127,343 episodes across 569 distinct kitchen scenes, spanning 14,155 scene configurations. The training split consists of 456 scenes (102,687 episodes), and the test split uses 113 fully held-out scenes (24,656 episodes). The scene-level split ensures that the test set evaluates generalization to novel spatial layouts, object arrangements, and clutter patterns.

Table 1. **Dataset Split Statistics.**

Split	Scenes	Configurations	Episodes
Train	456	$11,408$	$102,687$
Test (Unseen Scenes)	113	$2,747$	$24,656$

The annotation pipeline is fully automated: no human annotators are involved. Labels are deterministic and reproducible given the same scene seed and robot configuration.

The distribution of affordance scores across episodes reveals systematic variation with robot morphology. Arms with shorter reach (e.g., myCobot, 0.33 m) yield narrow feasibility bands close to the target, while long-reach arms (e.g., UR5e, 1.03 m) admit a wider arc of valid base poses. Scene clutter also modulates map shape: densely cluttered scenes produce fragmented affordance regions, whereas open layouts yield broader, more contiguous feasibility zones, making the dataset appropriately challenging across its full range of configurations.

On average, 18.3% of floor points per episode receive affordance scores above 0.5, with a standard deviation of 12.7% across episodes. Articulated targets (*e.g.*, drawers and cabinet doors) yield higher positive rates (21.5%) than rigid objects (16.2%), as they afford approach from multiple angles. This systematic variation underscores why continuous, densified labels are preferable to binary annotations: sparse binary labels would provide inadequate gradient signal in heavily cluttered episodes where feasible base poses are rare. Furthermore, the benchmark exhibits a long-tail distribution of feasibility region sizes: the smallest decile of episodes contains feasibility zones covering under 5% of floor points, providing targeted training signal for the most constrained manipulation scenarios that real-world deployments frequently encounter.

4. The NavAff Baseline

4.1. Model Overview

NavAff takes as input: (i) a global scene point cloud $\mathcal{P}_g \in \mathbb{R}^{N_g \times 4}$ with an additional feature channel encoding target and obstacle labels, (ii) a floor point cloud $\mathcal{P}_f \in \mathbb{R}^{N_f \times 3}$, and (iii) a robot descriptor $\mathbf{r} \in \mathbb{R}^2$ (base height, arm reach). It outputs per-point affordance scores $\hat{\mathcal{A}} \in [0, 1]^{N_f}$.

4.2. Point Cloud Encoders

Both the global and floor point clouds are encoded using PointNet++ [22] with Multi-Scale Grouping (MSG). The encoder consists of three set abstraction layers:

- **SA1**: 512 centroids, radii $[0.1, 0.2, 0.4]$, MLP channels $[[32, 32, 64], [64, 64, 128], [64, 96, 128]]$

AI-Generated

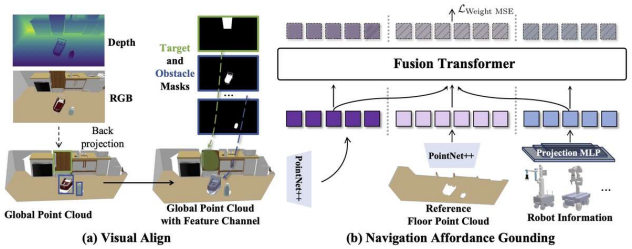

Figure 4. **NavAff Baseline.** NavAff encodes the global point cloud (with target and obstacle feature channels) and the floor point cloud via separate PointNet++ encoders, fuses them through a cross-attention Visual Affordance Module (VAM), and decodes per-floor-point affordance scores conditioned on a robot MLP embedding. The output is a continuous affordance map over the floor point cloud.

- **SA2**: 128 centroids, radii $[0.4, 0.8]$, MLP channels $[[128, 128, 256], [128, 196, 256]]$
- **SA3**: $N_p = 64$ centroids, radii $[0.2, 0.4]$, MLP channels $[[128, 128, 256], [128, 196, 256]]$

The global encoder receives the 4-channel input (XYZ + feature), while the floor encoder operates on raw XYZ coordinates.

4.3. Visual Affordance Module (VAM)

The Visual Affordance Module is a cross-attention layer [29] that fuses information from the global scene into the floor features:

$$\mathbf{F}'_f = \text{CrossAttn}(\mathbf{F}_f,\ [\mathbf{e}_r;\ \mathbf{F}_g]), \tag{2}$$

where $\mathbf{F}_f \in \mathbb{R}^{N_p \times d}$ are the floor features at the deepest scale, $\mathbf{F}_g \in \mathbb{R}^{N_p \times d}$ are the corresponding global features, and $\mathbf{e}_r \in \mathbb{R}^{1 \times d}$ is the robot embedding produced by a two-layer MLP:

$$\mathbf{e}_r = \text{MLP}(\mathbf{r}) = W_2\, \text{ReLU}(W_1\, \mathbf{r}), \tag{3}$$

with $W_1 \in \mathbb{R}^{256 \times 2}$ and $W_2 \in \mathbb{R}^{512 \times 256}$. The concatenation $[\mathbf{e}_r;\ \mathbf{F}_g]$ passes as keys and values to the cross-attention, so the floor queries attend jointly to robot morphology and scene context.

4.4. Decoder

Fused floor features are upsampled through three Feature Propagation (FP) layers back to the original floor point count N_f, following the PointNet++ decoder design. The final per-point affordance score is produced by a small MLP head followed by sigmoid activation. The entire network is trained end-to-end using a binary cross-entropy loss between predicted and ground-truth affordance maps.

4.5. Training Details

NavAff is trained for 100 epochs on the training split using the Adam optimizer [12] with an initial learning rate of 10^{-3} and cosine annealing. Training takes approximately 12 hours on a single NVIDIA A100 GPU. Floor and global point clouds are normalized to zero mean and unit scale independently. The floor point count is fixed at $N_f = 10{,}701$ (zero-padded as necessary) and the global count at $N_g = 16{,}647$.

5. Experiments

5.1. Evaluation Metrics

We evaluate affordance prediction quality using four metrics:

- **RMSE**: Root-Mean-Squared Error between predicted and ground-truth affordance maps.
- **logMSE**: Mean-Squared Error in log space, emphasizing small-value accuracy.
- **PCC**: Pearson Correlation Coefficient, measuring linear correlation.
- **SIM**: Cosine similarity between predicted and ground-truth affordance vectors.

We additionally report **Manipulation Success Rate (MSR)** to assess end-to-end utility. Given a predicted affordance map, we select the floor point with the highest predicted score (`Top1`) or the best among the top-5 scoring points (`Top5`) as the proposed base pose, navigate the robot there, and attempt manipulation. MSR is the fraction of episodes where manipulation succeeds.

5.2. Baselines

We compare NavAff against three adapted 3D point cloud methods:

- **PointNet++** [22]: A hierarchical point set network adapted to regress per-floor-point affordance scores from the global point cloud.
- **VoteNet** [23]: A Hough-voting-based 3D detection network adapted for floor affordance regression.
- **H3DNet** [44]: A hybrid geometric primitive network adapted for the same task.
- **Random**: A baseline that selects a uniformly random floor point, providing a lower bound on MSR.

All baselines receive the same global point cloud input as NavAff.

5.3. Main Results

Table 2. **Main results.** Quantitative evaluation of navigation affordance grounding performance of NavAff.

Method	RMSE ↓	logMSE ↓	PCC ↑	SIM ↑
PointNet++ [22]	0.164	0.0142	0.565	0.589
VoteNet [23]	0.167	0.0143	0.543	0.570
H3DNet [44]	0.174	0.0156	0.503	0.522
NavAff	**0.147**	**0.0115**	**0.680**	**0.696**

Table 2 reports quantitative results on the test split.

AI-Generated

Table 3. **Manipulation Success Rate (MSR)**.

MSR	Random	H3DNet	VoteNet	PointNet++	NavAff
Top1	0.080	0.54	0.56	0.60	**0.72**
Top5	0.046	0.47	0.53	0.58	**0.66**

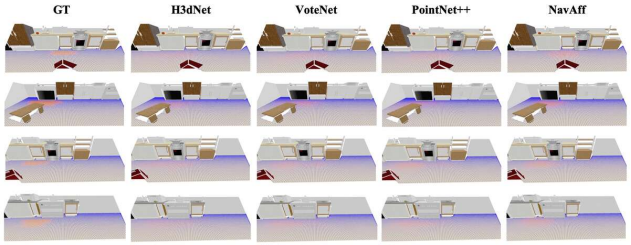

Figure 5. **Qualitative comparison of navigation affordance between all methods and ground truth.** NavAff (rightmost) produces affordance maps most closely aligned with the ground truth (second from left), accurately capturing the feasible manipulation region while suppressing erroneous predictions near obstacles and kinematically infeasible zones.

NavAff achieves the best performance across all four metrics, with RMSE of 0.147, logMSE of 0.0115, PCC of 0.680, and SIM of 0.696. Compared to the strongest baseline PointNet++, NavAff reduces RMSE by 10.4% (0.164→0.147), and improves PCC by 20.4% (0.565→0.680). H3DNet performs worst, likely because its bounding-box-centric representation is ill-suited for the dense floor regression task.

Table 3 shows MSR results. NavAff achieves a Top-1 MSR of 0.72, substantially outperforming PointNet++ (0.60), VoteNet (0.56), H3DNet (0.54), and the random baseline (0.08). The Top-5 MSR of NavAff is 0.66, also exceeding all baselines, demonstrating that the top-ranked predictions of NavAff are consistently reliable.

Notably, the gap between Top-1 and Top-5 MSR for NavAff (0.06) is smaller than for the baselines (0.08–0.14), indicating that its high-confidence predictions are consistently reliable rather than occasionally correct by chance. This is practically important: a robot that can trust its top-ranked pose avoids costly re-planning or backtracking when the first attempt fails, reducing overall task execution time.

Figure 5 shows qualitative comparisons. NavAff produces smooth, concentrated affordance maps that closely match the ground truth, correctly localizing the feasible manipulation zone. Baseline methods produce noisier or poorly localized predictions; H3DNet in particular generates diffuse maps that fail to identify the optimal approach region.

Per-Category Performance. Beyond aggregate metrics, we examine prediction quality broken down by object category. Articulated targets (*e.g.*, drawers, cabinet doors) achieve a higher average PCC of 0.704, whereas rigid objects (*e.g.*, mugs, bottles) score 0.651. We attribute this gap to the physically constrained approach axes of articulated objects: drawer handles require frontal access, cabinet doors require lateral clearance, producing structured affordance patterns that are easier to regress from geometric cues. Rigid objects admit approach from multiple directions, yielding broader but less distinctive affordance gradients. Thin cylindrical objects (bottles, cans) are particularly challenging—their small cross-section means a few centimeters of lateral displacement can shift the outcome from success to failure, demanding fine-grained spatial discrimination. Obstacle density further amplifies this difficulty: in the most cluttered quartile of test scenes, PCC for rigid objects drops to 0.618 for NavAff and 0.511 for the best baseline, underscoring the importance of the global point cloud for resolving occlusion patterns around compact targets.

Table 4. **Ablation study results of NavAff.**

	RMSE ↓	logMSE ↓	PCC ↑	SIM ↑	top1MSR ↑
NavAff	0.147	0.0115	0.680	0.696	0.72
w/o robot information	0.148	0.0115	0.670	0.688	0.70
w/o VAM	0.165	0.0140	0.562	0.589	0.63
w/o global point cloud	0.168	0.0144	0.534	0.568	0.58

5.4. Ablation Study

Table 4 presents ablation results on the test set. We study three ablations:

w/o robot information. Removing the robot descriptor $\mathbf{r}$ yields a small degradation (PCC: 0.680→0.670, Top-1 MSR: 0.72→0.70), suggesting that explicit robot-aware conditioning contributes to performance, especially for arms with unusual reach profiles.

w/o VAM. Removing the cross-attention Visual Affordance Module (replacing it with a direct concatenation) causes a substantial drop (PCC: 0.680→0.562, Top-1 MSR: 0.72→0.63), demonstrating that cross-attention-based fusion of global scene context into floor queries is critical for accurate affordance prediction.

w/o global point cloud. Removing the global point cloud entirely (i.e., predicting from floor points alone) causes the largest degradation (PCC: 0.680→0.534, Top-1 MSR: 0.72→0.58), confirming that information about the full scene—including target identity, obstacle layout, and furniture configuration—is essential for predicting feasible base positions.

5.5. Per-Robot Analysis

Performance varies systematically with robot morphology. Arms with shorter maximum reach (myCobot: 0.33 m) produce narrow feasibility bands that are harder to predict (PCC: 0.641), while long-reach arms (UR5e: 1.03 m) exhibit broader feasibility zones and achieve PCC of 0.712. Base height further modulates difficulty: taller bases restrict reachable object heights, sharpening feasibility boundaries and improving prediction confidence. These results motivate

AI-Generated

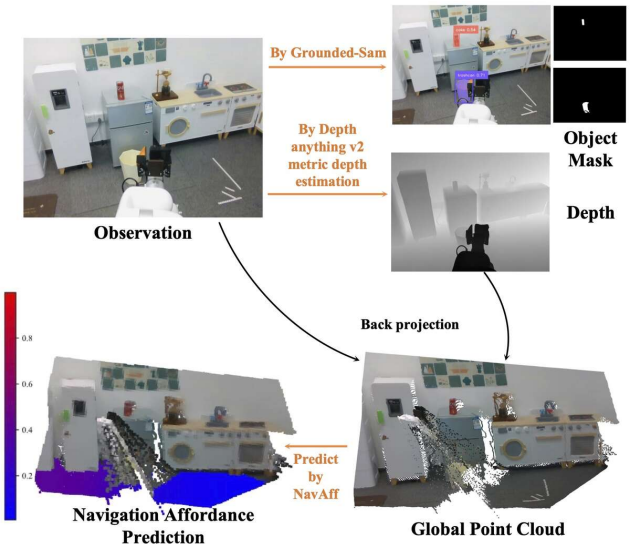

Figure 6. **Pipeline of real-world application with simulator-trained NavAff.** An Intel RealSense D435i provides RGB-D input; Grounded SAM [27] segments the target object and obstacles. The resulting point clouds are fed to the simulator-trained NavAff, which predicts floor affordances overlaid on the real kitchen floor.

richer kinematic representations beyond the two-dimensional descriptor used in NavAff. The robot descriptor embedding generalizes more effectively between morphologically similar arms (*e.g.*, Panda and Flexiv, whose reaches differ by only 0.07 m) than between extremes such as myCobot and UR5e, suggesting that incorporating joint limits and workspace volume into the robot encoding could further improve cross-morphology transfer.

We additionally observe that prediction error is concentrated near the boundaries of feasibility regions—precisely where the affordance score transitions from high to low. This boundary sensitivity is amplified for robots with sharp kinematic cutoffs, such as the Realman arm, whose rigid wrist limits create abrupt reachability boundaries in the floor plane. Addressing this boundary ambiguity may require supervision at finer spatial resolution or a curriculum training strategy that upweights floor points near affordance boundaries, where accurate localization is most critical for reliable base pose selection.

5.6. Sim-to-Real Transfer

Figure 6 illustrates our sim-to-real pipeline. We deploy a simulator-trained NavAff in real kitchen environments using an Intel RealSense D435i RGB-D camera mounted on the robot arm. Target and obstacle point clouds are obtained via Grounded SAM [27] open-vocabulary segmentation. The floor point cloud is extracted by thresholding estimated depth at low heights. The simulator-trained model produces visually plausible floor affordances in the real environment, correctly highlighting manipulable regions near the target and suppressing infeasible zones near obstacles, demonstrating zero-shot sim-to-real transfer.

To quantify real-world fidelity, we collected 15 real kitchen configurations with known target placements and manually verified whether the top-1 predicted base pose enabled a successful grasp. The simulator-trained NavAff achieved a top-1 MSR of 0.67 in these trials, a modest reduction from the simulated 0.72 attributable to RealSense depth noise and imperfect open-vocabulary segmentation. Failure cases were concentrated in scenes with transparent objects (*e.g.*, glass bottles) or highly reflective surfaces, where reliable depth estimation is difficult. These results confirm that geometric point-cloud representations exhibit a smaller sim-to-real domain gap than appearance-based features, and they motivate future work on noise-robust depth preprocessing and domain adaptation strategies tailored to common kitchen materials.

6. Conclusion

We have presented MoMa-Kitchen, a large-scale benchmark for affordance-grounded last-mile navigation in mobile manipulation, comprising 127,343 episodes with dense floor affordance maps across 569 procedurally generated kitchen scenes and six robot morphologies. MoMa-Kitchen operationalizes the critical but underexplored problem of predicting robot base poses that enable feasible manipulation, providing scalable, fully automated supervision for this task. We introduced NavAff, a lightweight baseline combining hierarchical point cloud encoding with cross-attention-based global–floor fusion and robot-aware conditioning, which outperforms adapted 3D detection baselines across all metrics. Ablation studies underscore the importance of the Visual Affordance Module and global point cloud context. Sim-to-real experiments demonstrate promising transfer to physical kitchen environments without fine-tuning.

Despite these promising results, MoMa-Kitchen currently focuses on single-arm grasping in kitchen environments; extending to bimanual manipulation, deformable objects, and non-kitchen scenes is an important avenue for future work. The two-dimensional robot descriptor captures only coarse kinematic properties; incorporating full joint limits and workspace reachability maps could yield more precise affordance boundaries for morphologically complex arms. Another promising direction is closed-loop affordance refinement: as a robot navigates toward a target, re-estimating the floor affordance map from successive viewpoints could enable iterative base pose correction, reducing positioning errors in dynamic or partially observed environments where a single prediction from a fixed viewpoint may be unreliable.

We hope MoMa-Kitchen will catalyze research at the intersection of navigation and manipulation, providing a foundation for learning-based approaches to robust mobile

AI-Generated

manipulation in cluttered domestic environments. Future directions include incorporating richer robot kinematic models, extending to multi-step manipulation tasks, and scaling to larger, more diverse scene types.

Acknowledgements. We thank the anonymous reviewers for their constructive feedback.

References

[1] Peter Anderson, Angel X. Chang, Devendra Singh Chaplot, Alexey Dosovitskiy, Saurabh Gupta, Vladlen Koltun, Jana Kosecka, Jitendra Malik, Roozbeh Mottaghi, Manolis Savva, and Amir Roshan Zamir. On evaluation of embodied navigation agents. *arXiv preprint arXiv:1807.06757*, 2018. 1

[2] Peter Anderson, Qi Wu, Damien Teney, Jake Bruce, Mark Johnson, Niko Sünderhauf, Ian Reid, Stephen Gould, and Anton Van Den Hengel. Vision-and-language navigation: Interpreting visually-grounded navigation instructions in real environments. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3674–3683, 2018. 1, 2

[3] Peter Anderson, Qi Wu, Damien Teney, Jake Bruce, Mark Johnson, Niko Sünderhauf, Ian Reid, Stephen Gould, and Anton van den Hengel. Vision-and-language navigation: Interpreting visually-grounded navigation instructions in real environments. 2018. 2

[4] Dhruv Batra, Angel X Chang, Sonia Chernova, Andrew J Davison, Jia Deng, Vladlen Koltun, Sergey Levine, Jitendra Malik, Igor Mordatch, Roozbeh Mottaghi, et al. Rearrangement: A challenge for embodied ai. *arXiv preprint arXiv:2011.01975*, 2020. 2

[5] Angel Chang, Angela Dai, Thomas Funkhouser, Maciej Halber, Matthias Niessner, Manolis Savva, Shuran Song, Andy Zeng, and Yinda Zhang. Matterport3d: Learning from rgb-d data in indoor environments. *arXiv preprint arXiv:1709.06158*, 2017. 2

[6] Angel X Chang, Thomas Funkhouser, Leonidas Guibas, Pat Hanrahan, Qixing Huang, Zimo Li, Silvio Savarese, Manolis Savva, Shuran Song, Hao Su, et al. Shapenet: An information-rich 3d model repository. *arXiv preprint arXiv:1512.03012*, 2015. 3

[7] Abhishek Das, Samyak Datta, Georgia Gkioxari, Stefan Lee, Devi Parikh, and Dhruv Batra. Embodied question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1–10, 2018. 1, 2

[8] Zipeng Fu, Xuxin Cheng, and Deepak Pathak. Deep whole-body control: learning a unified policy for manipulation and locomotion. In *Conference on Robot Learning*, pages 138–149. PMLR, 2023. 2

[9] Zipeng Fu, Tony Z Zhao, and Chelsea Finn. Mobile aloha: Learning bimanual mobile manipulation with low-cost whole-body teleoperation. *arXiv preprint arXiv:2401.02117*, 2024. 1, 2

[10] Wenlong Huang, Chen Wang, Ruohan Zhang, Yunzhu Li, Jiajun Wu, and Li Fei-Fei. Voxposer: Composable 3d value maps for robotic manipulation with language models. *ArXiv*, abs/2307.05973, 2023. 2, 3

[11] Mukul Khanna, Yongsen Mao, Hanxiao Jiang, Sanjay Haresh, Brennan Shacklett, Dhruv Batra, Alexander Clegg, Eric Undersander, Angel X Chang, and Manolis Savva. Habitat synthetic scenes dataset (hssd-200): An analysis of 3d scene scale and realism tradeoffs for objectgoal navigation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16384–16393, 2024. 2

[12] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization, 2017. 5

[13] Jacob Krantz, Erik Wijmans, Arjun Majumdar, Dhruv Batra, and Stefan Lee. Beyond the nav-graph: Vision-and-language navigation in continuous environments. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXVIII 16*, pages 104–120. Springer, 2020. 1

[14] Jacob Krantz, Erik Wijmans, Arjun Majumdar, Dhruv Batra, and Stefan Lee. Beyond the nav-graph: Vision-and-language navigation in continuous environments. 2020. 2

[15] Alexander Ku, Peter Anderson, Roma Patel, Eugene Ie, and Jason Baldridge. Room-across-room: Multilingual vision-and-language navigation with dense spatiotemporal grounding. *arXiv preprint arXiv:2010.07954*, 2020. 2

[16] Chengshu Li, Fei Xia, Roberto Martin-Martin, and Silvio Savarese. Hrl4in: Hierarchical reinforcement learning for interactive navigation with mobile manipulators. In *Conference on Robot Learning*, pages 603–616. PMLR, 2020. 2

[17] Peiqi Liu, Yaswanth Orru, Jay Vakil, Chris Paxton, Nur Muhammad Mahi Shafiullah, and Lerrel Pinto. Ok-robot: What really matters in integrating open-knowledge models for robotics. *arXiv preprint arXiv:2401.12202*, 2024. 2

[18] Andrew Melnik, Michael Büttner, Leon Harz, Lyon Brown, Gora Chand Nandi, Arjun PS, Gaurav Kumar Yadav, Rahul Kala, and Robert Haschke. Uniteam: Open vocabulary mobile manipulation challenge. *arXiv preprint arXiv:2312.08611*, 2023. 2

[19] Karan Mirakhor, Sourav Ghosh, Dipanjan Das, and Brojeshwar Bhowmick. Task planning for visual room rearrangement under partial observability. In *The Twelfth International Conference on Learning Representations*, 2024. 2

[20] Mayank Mittal, David Hoeller, Farbod Farshidian, Marco Hutter, and Animesh Garg. Articulated object interaction in unknown scenes with whole-body mobile manipulation. In *2022 IEEE/RSJ international conference on intelligent robots and systems (IROS)*, pages 1647–1654. IEEE, 2022. 1, 2

[21] Kaichun Mo, Shilin Zhu, Angel X. Chang, Li Yi, Subarna Tripathi, Leonidas J. Guibas, and Hao Su. PartNet: A large-scale benchmark for fine-grained and hierarchical part-level 3D object understanding. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019. 2, 3

[22] Charles Ruizhongtai Qi, Li Yi, Hao Su, and Leonidas J Guibas. Pointnet++: Deep hierarchical feature learning on point sets in a metric space. *Advances in neural information processing systems*, 30, 2017. 3, 4, 5

[23] Charles R. Qi, Or Litany, Kaiming He, and Leonidas J. Guibas. Deep hough voting for 3d object detection in point clouds. In

AI-Generated

Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2019. 3, 5

[24] Yuankai Qi, Qi Wu, Peter Anderson, Xin Wang, William Yang Wang, Chunhua Shen, and Anton van den Hengel. Reverie: Remote embodied visual referring expression in real indoor environments. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9982–9991, 2020. 1, 2

[25] Dicong Qiu, Wenzong Ma, Zhenfu Pan, Hui Xiong, and Junwei Liang. Open-vocabulary mobile manipulation in unseen dynamic environments with 3d semantic maps. *arXiv preprint arXiv:2406.18115*, 2024. 2

[26] Santhosh K Ramakrishnan, Aaron Gokaslan, Erik Wijmans, Oleksandr Maksymets, Alex Clegg, John Turner, Eric Undersander, Wojciech Galuba, Andrew Westbury, Angel X Chang, et al. Habitat-matterport 3d dataset (hm3d): 1000 large-scale 3d environments for embodied ai. *arXiv preprint arXiv:2109.08238*, 2021. 2

[27] Tianhe Ren, Shilong Liu, Ailing Zeng, Jing Lin, Kunchang Li, He Cao, Jiayu Chen, Xinyu Huang, Yukang Chen, Feng Yan, Zhaoyang Zeng, Hao Zhang, Feng Li, Jie Yang, Hongyang Li, Qing Jiang, and Lei Zhang. Grounded sam: Assembling open-world models for diverse visual tasks. *ArXiv*, abs/2401.14159, 2024. 2, 3, 7

[28] Beichen Shao, Nieqing Cao, Yan Ding, Xingchen Wang, Fuqiang Gu, and Chao Chen. Moma-pos: An efficient object-kinematic-aware base placement optimization framework for mobile manipulation. *arXiv preprint arXiv:2403.19940*, 2024. 2

[29] A Vaswani. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017. 3, 5

[30] Weikang Wan, Haoran Geng, Yun Liu, Zikang Shan, Yaodong Yang, Li Yi, and He Wang. Unidexgrasp++: Improving dexterous grasping policy learning via geometry-aware curriculum and iterative generalist-specialist learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 3891–3902, 2023. 2

[31] Maozhen Wang, Rui Luo, Aykut Özgün Önol, and Taşkin Padir. Affordance-based mobile robot navigation among movable obstacles. In *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 2734–2740. IEEE, 2020. 2

[32] Ruicheng Wang, Jialiang Zhang, Jiayi Chen, Yinzhen Xu, Puhao Li, Tengyu Liu, and He Wang. Dexgraspnet: A large-scale robotic dexterous grasp dataset for general objects based on simulation. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pages 11359–11366. IEEE, 2023. 2

[33] Luca Weihs, Matt Deitke, Aniruddha Kembhavi, and Roozbeh Mottaghi. Visual room rearrangement. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5922–5931, 2021. 2

[34] Erik Wijmans, Samyak Datta, Oleksandr Maksymets, Abhishek Das, Georgia Gkioxari, Stefan Lee, Irfan Essa, Devi Parikh, and Dhruv Batra. Embodied question answering in photorealistic environments with point cloud perception. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6659–6668, 2019. 2

[35] Josiah Wong, Albert Tung, Andrey Kurenkov, Ajay Mandlekar, Li Fei-Fei, Silvio Savarese, and Roberto Martín-Martín. Error-aware imitation learning from teleoperation data for mobile manipulation. In *Conference on Robot Learning*, pages 1367–1378. PMLR, 2022. 2

[36] Bohan Wu, Roberto Martin-Martin, and Li Fei-Fei. M-ember: Tackling long-horizon mobile manipulation via factorized domain transfer. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pages 11690–11697. IEEE, 2023. 2

[37] Fei Xia, Amir R Zamir, Zhiyang He, Alexander Sax, Jitendra Malik, and Silvio Savarese. Gibson env: Real-world perception for embodied agents. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 9068–9079, 2018. 2

[38] Fanbo Xiang, Yuzhe Qin, Kaichun Mo, Yikuan Xia, Hao Zhu, Fangchen Liu, Minghua Liu, Hanxiao Jiang, Yifu Yuan, He Wang, Li Yi, Angel X. Chang, Leonidas J. Guibas, and Hao Su. SAPIEN: A simulated part-based interactive environment. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020. 3

[39] Yinzhen Xu, Weikang Wan, Jialiang Zhang, Haoran Liu, Zikang Shan, Hao Shen, Ruicheng Wang, Haoran Geng, Yijia Weng, Jiayi Chen, et al. Unidexgrasp: Universal robotic dexterous grasping via learning diverse proposal generation and goal-conditioned policy. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4737–4746, 2023. 2

[40] Kui Yang, Nieqing Cao, Yan Ding, and Chao Chen. Bestman: A modular mobile manipulator platform for embodied ai with unified simulation-hardware apis. 2024. 1, 3

[41] Yuhang Yang, Wei Zhai, Hongchen Luo, Yang Cao, Jiebo Luo, and Zheng-Jun Zha. Grounding 3d object affordance from 2d interactions in images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10905–10915, 2023. 2

[42] Sriram Yenamandra, Arun Ramachandran, Karmesh Yadav, Austin Wang, Mukul Khanna, Theophile Gervet, Tsung-Yen Yang, Vidhi Jain, Alexander William Clegg, John Turner, et al. Homerobot: Open-vocabulary mobile manipulation. *arXiv preprint arXiv:2306.11565*, 2023. 1, 2

[43] Naoki Yokoyama, Alex Clegg, Joanne Truong, Eric Undersander, Tsung-Yen Yang, Sergio Arnaud, Sehoon Ha, Dhruv Batra, and Akshara Rai. Asc: Adaptive skill coordination for robotic mobile manipulation. *IEEE Robotics and Automation Letters*, 9(1):779–786, 2023. 2

[44] Zaiwei Zhang, Bo Sun, Haitao Yang, and Qi-Xing Huang. H3dnet: 3d object detection using hybrid geometric primitives. In *European Conference on Computer Vision*, 2020. 3, 5

[45] Peiyuan Zhi, Zhiyuan Zhang, Muzhi Han, Zeyu Zhang, Zhitian Li, Ziyuan Jiao, Baoxiong Jia, and Siyuan Huang. Closed-loop open-vocabulary mobile manipulation with gpt-4v. *arXiv preprint arXiv:2404.10220*, 2024. 2

[46] Jun Zhu, Zihao Du, Haotian Xu, Fengbo Lan, Zilong Zheng, Bo Ma, Shengjie Wang, and Tao Zhang. Navi2gaze: Leveraging foundation models for navigation and target gazing. *arXiv preprint arXiv:2407.09053*, 2024. 2

AI-Generated